

Diamorphism is a phenomenon related to geomorphology. Even while internal forces keep the crust solid, there is constant stirring and shifting occurring in some parts of the crust. Significantly earth movement, refers to the slow, abrupt or quick changes in the earth's crust caused by earthquakes or other motion influenced by internal forces. These include abrupt movement and gradual or diastrophic movements. Heterogenic processes do not come under diastrophism.

16. Consider the following :

1. Electromagnetic radiation
2. Geothermal energy
3. Gravitational force
4. Plate Movements
5. Rotation of Earth
6. Revolution of the Earth

Which of the following above are responsible for bringing dynamic changes on the surface of the Earth ?

- (a) 1,2,3 and 4
- (b) 1,3,5 and 6
- (c) 2,4,5 and 6
- (d) 1,2,3,4,5 and 6

I.A.S. (Pre) 2013

Ans. (d)

The dynamic changes on the Earth's surface are caused by endogenic forces. In general, the origin of endogenic force is due to thermal asymmetry in the inner part of the Earth and compression and expansion of rocks. In addition to these electromagnetic radiation, gravitational force, rotation of Earth and its revolution functions are also liable for changes on the Earth's surface.

17. Arrange in chronological order :

1. Patpara formation
2. Khetaunhi formation
3. Baghor formation
4. Sihawal formation

Codes :

- (a) 1, 4, 2, 3
- (b) 4, 1, 3, 2
- (c) 1, 2, 3, 4
- (d) 4, 3, 2, 1

U.P. P.C.S. (Mains) 2017

Ans. (b)

(Formation)	(Period)
Sihawal	Lower Palaeolithic
Patpara	Middle Palaeolithic
Baghor	Upper Palaeolithic
Khetaunhi	Neolithic and Microlithic age

18. Consider the following events.

1. First reptiles
2. First insects
3. Shelled animals
4. First mammals

Arrange the above events in Chronological order of their origin on the earth and select the correct answer from the codes given below :

Code :

- (a) 2 1 3 4
- (b) 2 3 1 4
- (c) 3 2 1 4
- (d) 2 4 1 3

U.P. P.C.S. (Pre) 2018

Ans. (*)

The right chronology is –

First reptiles and
First insects,
Shelled animals,
First mammals.

19. Rajasthan desert or Thar desert is the expanse of which of the following?

- (a) Pliocene
- (b) Paleocene
- (c) Pleistocene and recent deposits
- (d) Oligocene

U.P. P.C.S. (Pre) 2018

Ans. (c)

Rajasthan desert or Thar desert is the expanse of Pleistocene and recent deposits. It is a rainfall deficient. According to the Department of Archaeology, the formation of the Thar desert started approx 70,000 years ago, while according to NCERT, the 'Pleistocene' era lasted from 2 million years ago to 10,000 years ago.

Rocks

*The crust of the Earth is made up of rocks. Rock is a natural substance, a solid aggregate of one or more minerals or mineraloids. Petrology is the study of rocks, which is an essential component of geology.

*The major groups of rocks are defined as igneous, sedimentary and metamorphic.

Igneous Rocks – Igneous rock is formed through the cooling and solidification of magma or lava. They are granular

in texture and are not found in the stratified form. They are devoid of any kind of fossils. Igneous rocks often have a correlation with volcanic activities. Igneous rocks are hard for water to penetrate. Igneous rock is found both in crystalline and non-crystalline form. They are also known as primary rocks. Igneous rocks are divided into two main categories-

- (a) **Plutonic or intrusive rocks** – formed when magma cools and crystallises slowly within the earth's crust. Ex.-granite, gabbros and pegmatite.
- (b) **Volcanic or extrusive rocks are** – Basalt, volcanic breccias etc.

Sedimentary rocks – Sedimentary rocks are formed at the Earth's surface by the accumulation and cementation of fragments of earlier rocks, minerals and organisms or as chemical precipitates and organic growth in water (sedimentation).

*Sediments are formed by the weathering of earlier rocks (igneous, sedimentary and metamorphic) by erosion from a source and are then transported to the place of deposition by water, wind, ice and mass movement of glaciers (agents of denudation). After this through the process of lithification, loose, unconsolidated sediments are turned into solid sedimentary rocks.

*Many sedimentary rocks are able to retain their specific characteristics in their depository layer, even after lithification. It is the reason why sand stones have multiple layers of different concentrations.

*Sedimentary rocks are formed by the accumulation of sediments. There are three basic types of sedimentary rocks-

- Sedimentary rocks formed as a result of mechanical processes- Ex - limestone, sandstones, shale, conglomerate etc.
- sedimentary rocks - geyserite, limestone, coal, etc.
- Chemical sedimentary rocks - halite, potassium salt etc.

Metamorphic Rocks – These rocks are formed by subjecting any rock type – sedimentary rock or igneous rock to different temperature and pressure conditions and chemical processes than those in which the original rock was formed. This process is called metamorphism meaning “**change of form**”. The result is a profound change in physical properties and chemistry of the stone. The temperature and pressure required for this process are always higher than those found at the Earth's surface.

Metamorphosed form of Igneous Rocks –

- Granite – Gneiss
- Basalt – Schist

(3) Basalt – Amphibolite
Metamorphosed form of Sedimentary Rocks -

- Shale – Slate
- Limestone – Marble
- Chalk and Dolomite – Marble
- Sandstone – Quartzite
- Conglomerate - Gneiss

Metamorphosed form of Metamorphic Rocks –

- Slate – Phyllite
- Phyllite – Schist
- Gabbro – Serpentine

1. Which type of rock is formed from cooling and solidification of lava ?

- | | |
|-------------|-----------------|
| (a) Igneous | (b) Sedimentary |
| (c) Crystal | (d) Metamorphic |

70th B.P.S.C. Pre-2024

Ans. (a)

Igneous rocks are formed by the cooling and solidification of magma/lava. Igneous rocks are of two types: intrusive and extrusive.

1. Extrusive Igneous Rocks: When magma comes out from the **Earth's interior** (e.g., volcanic eruptions) and cools and solidifies on the Earth's surface, it forms extrusive igneous rocks. An example is basalt rock.

2. Intrusive Igneous Rocks: When magma solidifies within the Earth's crust, it forms intrusive igneous rocks. An example is granite rock.

2. Consider the following statements about the sedimentary rocks:

- Sedimentary rocks are formed at the earth's surface by the hydrological system**
- The formation of sedimentary rocks involves the weathering of preexisting rocks**
- Sedimentary rocks contain fossils**
- Sedimentary rocks typically occur in layers**

Which of these statements are correct?

- | | |
|----------------|-------------------|
| (a) 1 and 2 | (b) 1 and 4 |
| (c) 2, 3 and 4 | (d) 1, 2, 3 and 4 |

I.A.S. (Pre) 2001

Ans. (d)

Sedimentary rocks are formed by the accumulation of sediments. This sediment or debris accumulates in low lying areas like-lakes. These sedimentary materials may be formed from eroded fragments of other or pre-existing rocks or even from the remains of plants or animals and fossils, most frequently found in sedimentary rocks, which comes in layers, called strata. Thus all of the given statements are true related to sedimentary rocks.

3. Regarding sedimentary rocks which of the following statements is true?

- (a) These are such rocks whose structure depend on temperature and pressure
- (b) These rocks are crystal
- (c) These rocks are deposited in layers
- (d) These rocks cannot be formed in water

39th B.P.S.C. (Pre) 1994

Ans. (c)

See the explanation of the above question.

4. Sandstone is flaky rock, because –

- (a) It is formed in the desert
- (b) It is formed due to Heat
- (c) It is formed underwater
- (d) It is formed on the mountain

M.P.P.C.S. (Pre) 1993

Ans. (c)

Sandstone is a sedimentary rock composed of sand-sized grains of mineral, rock or organic materials. The sandstone forms under the sea or in low-lying areas. It converts into quartzite, which is typically the most abundant type of sand grain present in sandstone. Chert, conglomerate, flints, etc. are some important types of sedimentary rock.

5. Which statement of the following is true for igneous rocks?

- (a) These have little fossils
- (b) They have porous for water
- (c) They are both crystal and non-crystal
- (d) These rocks have no silicas

40th B.P.S.C. (Pre) 1995

Ans. (c)

The origin of igneous rocks is mainly related to the process of a volcanic eruption. These types of rocks are formed by the cooling and solidification of material called magma or lava. Igneous rocks have both, crystalline and non-crystalline structures and thus the statement given in option (c) is true but all other given statements are wrong because these rocks are composed of silicate minerals and fossils and layers are not found in these types of rocks. Additionally water supply through igneous rocks is very low.

6. Which of the following rocks is not likely to contain fossils?

- (a) Conglomerate
- (b) Granite
- (c) Shale
- (d) Sandstone

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

Among the given options, granite does not contain fossils. Granite is formed from magma from a volcanic eruption. It is an example of Plutonic or Intrusive rocks. Conglomerate, sandstone and shale are examples of sedimentary rocks. Sedimentary rocks contain fossils.

7. Origin of Metamorphic rocks occurs from rock ?

- (a) Igneous
- (b) Sedimentary
- (c) Both Igneous and Sedimentary
- (d) None of the above

Jharkhand P.C.S. (Pre) 2011

Ans. (c)

"Metamorphism" is a Greek word which means "change of form". Metamorphic rock is a rock that has changed from one type of rock into another. As a result of intense heat from magma and pressure from tectonic shifting, these metamorphic rocks are produced from either igneous rocks or sedimentary rocks.

8. Which one of the following is not an example of Metamorphic rock ?

- (a) Marble
- (b) Quartzite
- (c) Slate
- (d) Granite

Uttarakhand P.C.S. (Pre) 2006

Ans. (d)

Granite is not an example of metamorphic rock. It is an igneous rock. The examples of metamorphic rocks formed from igneous rocks are as follows –

- (1) Granite - Gneiss
- (2) Basalt - Amphibolite
- (3) Basalt - Schist

Marble is an example of metamorphic rock formed from sedimentary rock. The other examples of metamorphic rocks formed from sedimentary rock are as follows-

- (1) Shale - Slate
- (2) Limestone - Marble
- (3) Chalk and dolomite - Marble
- (4) Sandstone - Quartzite

9. Which one of the following is not associated with the 'group'?

- (a) Gneiss
- (b) Sandstone
- (c) Limestone
- (d) Shale

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

In the given options Limestone, Sandstone and Shale belong to sedimentary rocks, while Gneiss is an example of metamorphic rock. Normally Gneiss is formed from granite.